Next factor-base experiment gates at subgroup order 2^{50}–2^{63}

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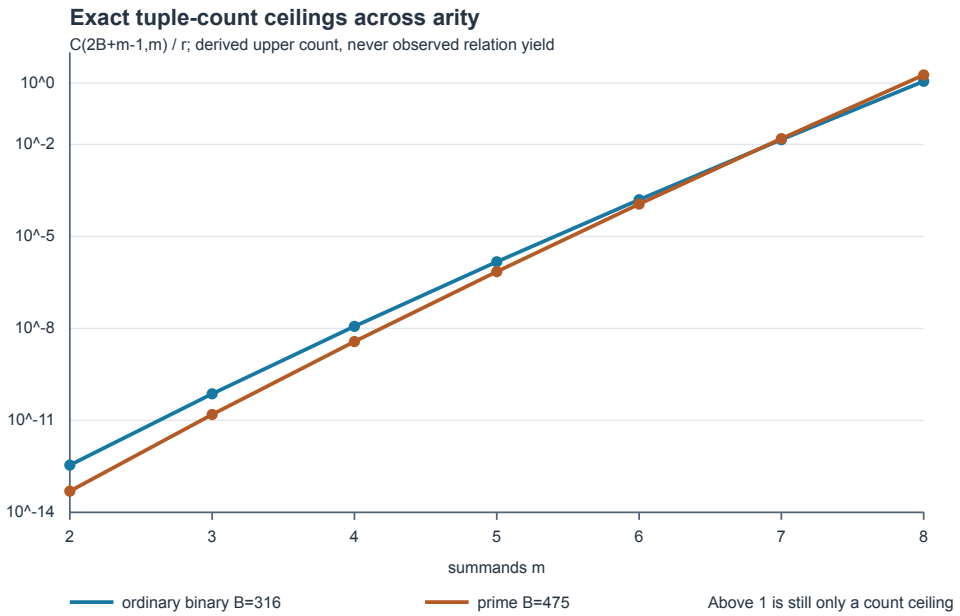
This round ranked prepared factor-base and solver interfaces for ordinary binary, prime, and Koblitz curves across $m = 2...8$. It used the frozen input and independent-replay records listed below. No scientific solver trial was run in this round; no end-to-end cost improvement follows from the prepared systems.

Ranked gates

Rank	Family	Prepared input	Next discriminating gate
1	Ordinary binary	GF(2^{61}), prime subgroup $r = 576460752557211763$. Two 316-column k11 bases, four matched channels, separately planted satisfiable formulas.	Frozen bounded CryptoMiniSat recovery of both controls, then a new common-target pilot with exact full-point checks.
2	Prime	$p = 9223372021248595841$, prime $r = 9223372018953067627$. Three equal 475-point bases; 21 common-target systems for $m = 2...8$.	Validate a genuine 63-bit polynomial backend against independent arithmetic and positive-root controls, then attempt matched capped systems.
3	Koblitz	Certified prime subgroups at field degrees 71/73; the reviewed GF(2^{53}) row targets a 45-bit subgroup.	Construct a target-blind cofactor-aware orbit base and same-column control, then a frozen pilot on one full-width fixture.

Exact count ceiling for the fixed prepared bases

For B projected columns and m signed terms, $\binom{2B+m-1}{m}$ counts signed multisets with repetition. The chart divides that count by the checked prime order r. Below one it bounds the representable fraction for uniformly sampled subgroup targets. Above one it is only a count ratio. It does not measure relation yield, point-sum collisions, algebraic regularity, or solver cost.



The fixed bases have count ratios below 0.016 through $m = 7$ and above one at $m = 8$. A complete $m = 2...8$ cost ladder needs arity-specific base sizes and must charge base construction, relation attempts, useful rank, linear algebra, target work, verification and matched rho. Short lower-arity panels on these particular bases cannot support a performance claim.

Evidence boundary and next check

The ordinary-binary positive witnesses satisfy every serialized CNF and XOR constraint, but no solver has recovered them. The prime common targets have public known-answer scalars; they are correctness fixtures, not planted decompositions. At prime $m = 8$, the coset direct input has 118 monomial occurrences versus 3910 and 3902 for equal-size controls; its quadratic lift has 103 variables and 104 equations. Those are input counts, not solving-degree or time observations. The Koblitz group fixtures are checked, but no complete full-width IC runner exists.

The immediate binary gate records solver and certificate viability under fixed caps; any modest base advantage requires a separate, powered target panel. The prime gate requires arithmetic-correct wide-field solving before timing. The Koblitz gate keeps field degree and prime subgroup order separate. The canonical ECDLP frontier graph was checked and needs no update because no reviewed full-cost result changed.

Evidence: docs/reports/2026-10-16-elite-factor-base-baseline.md; 2026-10-17-binary-k11-planted-xcnf.md; 2026-10-17-prime-475-target-systems.md; 2026-10-17-koblitz-size-screen.md; problems/full-width-ic-arity-grid-unmeasured.md. Chart source: 2026-10-17-icperf-next-experiments-chart.py. Numerical provenance: internal input records and derived exact binomial counts.